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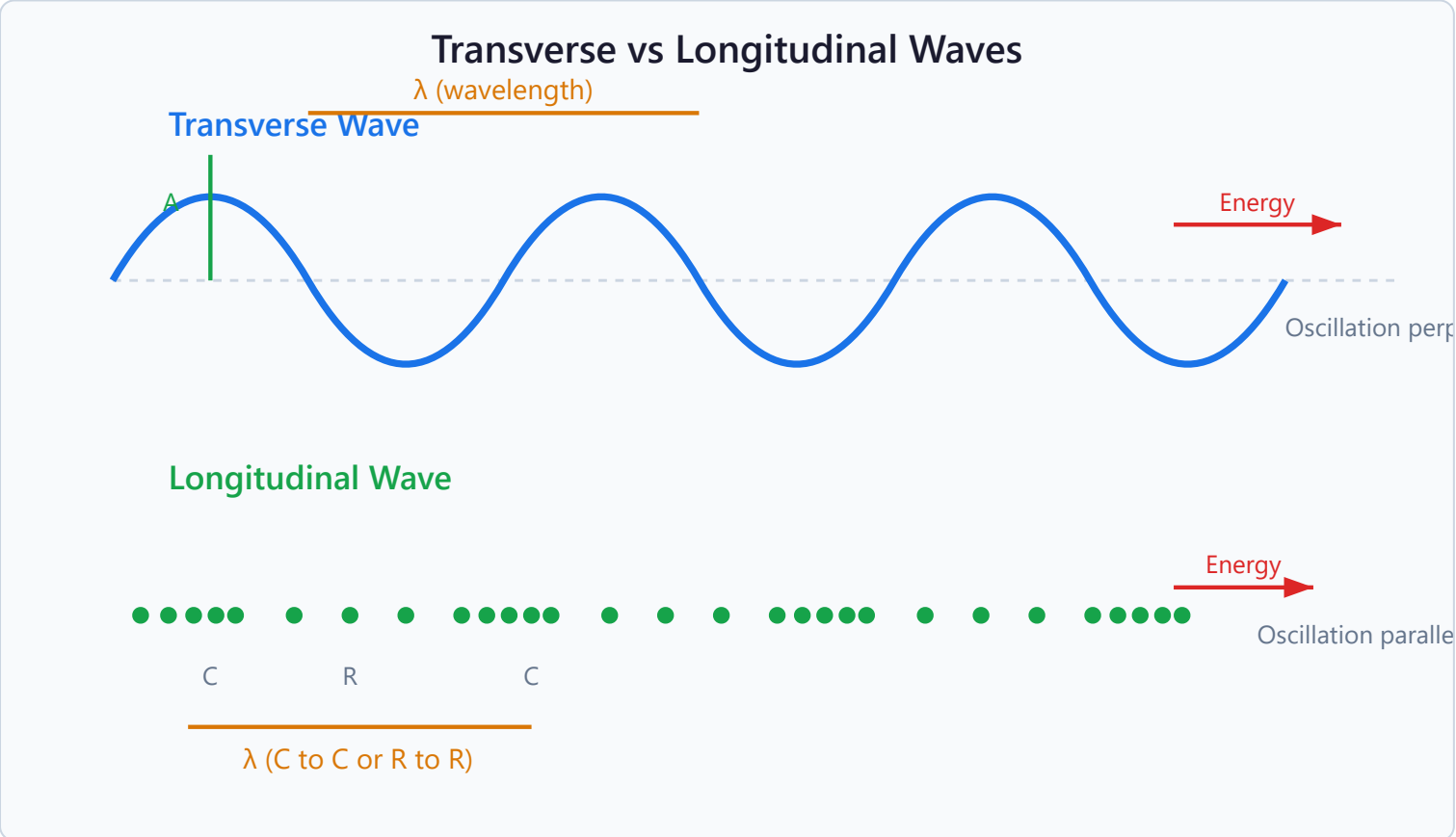
Topic 5 — Waves and Particle Nature of Light

5.1 Types of Waves

Transverse wave: Oscillations are perpendicular to the direction of energy transfer. Examples: light, all EM waves, water surface waves, waves on a string, S-waves (seismic).

Longitudinal wave: Oscillations are parallel to the direction of energy transfer. Examples: sound, ultrasound, P-waves (seismic), pressure waves in a slinky.

Longitudinal waves consist of **compressions** (high pressure/density regions) and **rarefactions** (low pressure/density regions). Only transverse waves can be polarised.



Polarisation

> **Polarisation:** Restricting the oscillations of a transverse wave to a single plane. Only transverse waves can be polarised (this is evidence that light is a transverse wave). Unpolarised light oscillates in all planes perpendicular to the direction of travel.

A **polarising filter** (Polaroid) only transmits waves oscillating in one plane. When two polarising filters are aligned, light passes through. When they are crossed (at 90 degrees), no light passes. This is called **Malus's Law**: $I = I_0\cos^2\theta$.

5.2 Wave Properties

Quantity	Definition	Symbol	Unit
Wavelength	Distance between two consecutive points in phase (crest to crest)	λ	m
Frequency	Number of complete oscillations per second	f	Hz (s^{-1})
Period	Time for one complete oscillation	T	s
Amplitude	Maximum displacement from the equilibrium position	A	m
Wave speed	Distance travelled by the wave per unit time	v	m s^{-1}

$$v = f\lambda \quad T = 1/f$$

The intensity of a wave is proportional to the square of the amplitude: $I \propto A^2$.

Phase and Phase Difference

> **Phase difference:** How much one wave is ahead of or behind another, measured in degrees (0-360) or radians (0-2π).

- **In phase:** Phase difference = 0 (or 2π, 4π, ...) = nλ path difference. Waves reinforce.

- **Antiphase:** Phase difference = π (or 3π, 5π, ...) = (n+½)λ path difference. Waves cancel.

$$\text{Phase difference} = (\Delta x / \lambda) \times 2\pi \text{ rad} = (\Delta x / \lambda) \times 360°$$

5.3 Superposition and Interference

Principle of superposition: When two or more waves overlap at a point, the resultant displacement equals the vector sum of the individual displacements at that point and at that instant.

Interference

- **Constructive:** Waves in phase (path difference = nλ). Amplitudes add. Bright fringe / loud sound.
- **Destructive:** Waves in antiphase (path difference = (n+½)λ). Amplitudes cancel. Dark fringe / silence.

Conditions for Observable Interference

- Sources must be **coherent** — same frequency and constant phase relationship.
- Waves must have similar amplitudes for complete cancellation.
- For transverse waves, they must be polarised in the same plane (or unpolarised).

Young's Double Slit Experiment

Coherent light passes through two narrow slits and produces an interference pattern of equally spaced bright and dark fringes on a screen.

$$w = \lambda D / d$$

where w = fringe spacing, λ = wavelength, D = slit-to-screen distance, d = slit separation.

Worked Example: In a Young's double slit experiment, a laser of wavelength 635 nm is used. The slit separation is 0.50 mm and the screen is 2.0 m away. Calculate the fringe spacing.

Solution:

$w = \lambda D/d = (635 \times 10^{-9} \times 2.0) / (0.50 \times 10^{-3}) = 1.27 \times 10^{-6} / 5.0 \times 10^{-4} =$ **2.54×10^{-3} m = 2.5 mm**

Exam Tip: Young's double slit experiment provides evidence for the wave nature of light. The experiment requires coherent sources (achieved by using a laser or by passing light through a single slit first to create a point source, then through two slits). With white light, the central fringe is white, but subsequent fringes show coloured edges because different wavelengths have different fringe spacings.

5.4 Stationary (Standing) Waves

A **stationary wave** is formed by the superposition of two progressive waves of equal frequency and amplitude travelling in opposite directions. Energy is stored but not transferred along the wave.

Key Features (compared to progressive waves)

Feature	Stationary Wave	Progressive Wave
Energy transfer	No net energy transfer	Energy transferred in direction of travel
Amplitude	Varies from 0 (node) to 2A (antinode)	Same for all points
Phase	All points between two nodes are in phase; points on opposite sides of a node are in antiphase	Phase varies continuously along the wave
Wavelength	Distance between alternate nodes = λ	Distance between successive identical points

Harmonics on a String (both ends fixed)

Harmonic	Antinodes	Nodes	Frequency	Wavelength
1st (fundamental)	1	2 (ends)	f_0	$\lambda = 2L$
2nd	2	3	$2f_0$	$\lambda = L$

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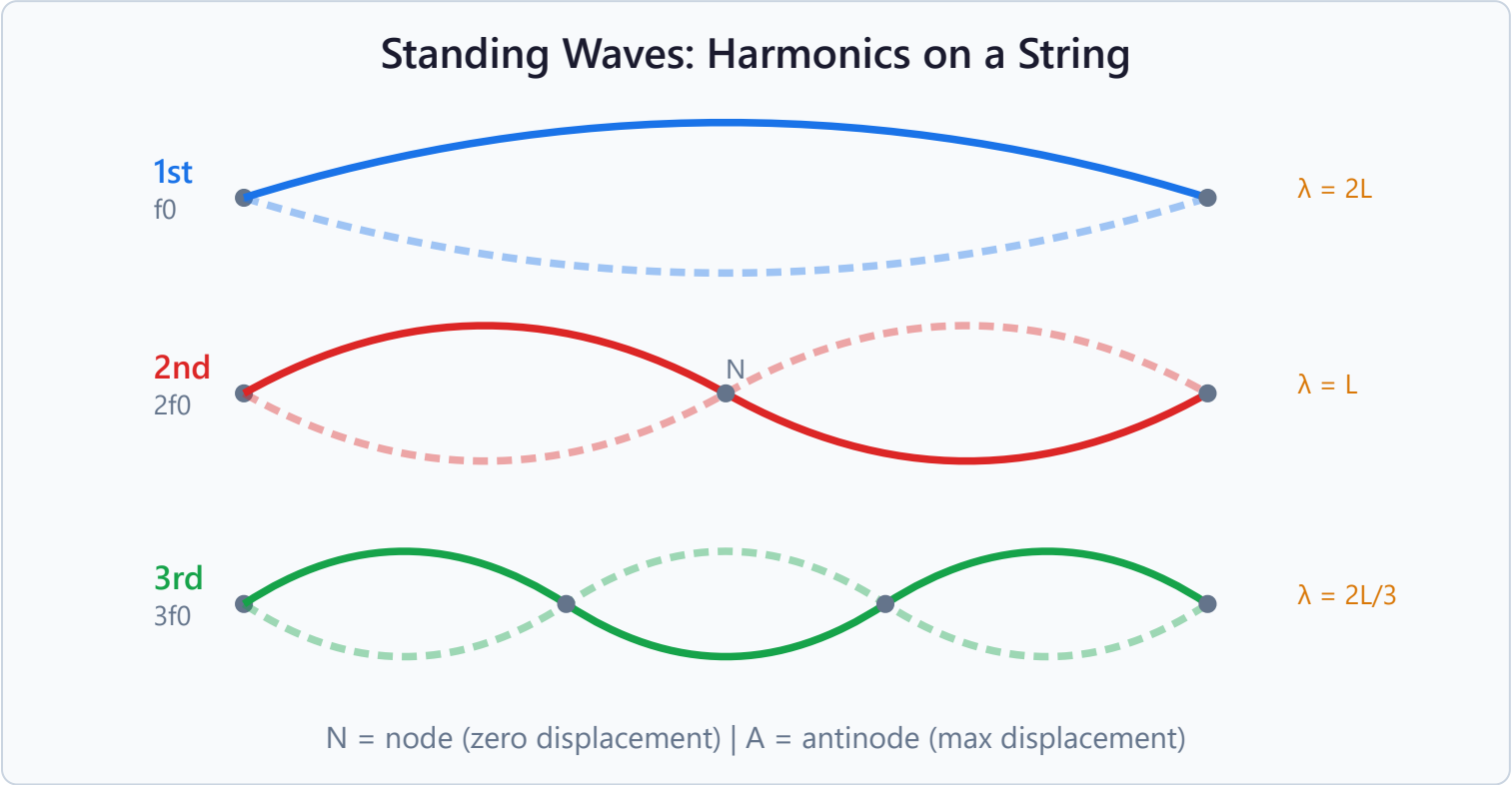
3rd	3	4	$3f_0$	λ = 2L/3
nth	n	n+1	nf_0	λ = 2L/n



Stationary Waves in Pipes

- **Open pipe (both ends open):** Antinodes at both ends. All harmonics (1st, 2nd, 3rd...). Fundamental: λ = 2L.
- **Closed pipe (one end closed):** Node at closed end, antinode at open end. Only odd harmonics (1st, 3rd, 5th...). Fundamental: λ = 4L.

5.5 Diffraction

Diffraction: The spreading of a wave as it passes through a gap or around an obstacle. Maximum diffraction occurs when the gap width is approximately equal to the wavelength.

Single Slit Diffraction

A single slit produces a central maximum that is twice the width of subsidiary maxima. Most of the intensity is in the central maximum. Wider slit = narrower central maximum (less diffraction).

Diffraction Grating

A grating with many equally spaced slits produces sharp, bright maxima at specific angles:

$$d \sin \theta = n\lambda$$

where d = slit spacing = 1/N (N = number of slits per metre), θ = angle of nth order maximum.

Worked Example: A diffraction grating has 500 lines per mm. Light of wavelength 590 nm is used. Calculate (a) the slit spacing, (b) the angle of the 2nd order maximum, (c) the maximum observable order.

Solution:

(a) $d = 1/(500 \times 10^3) = 1/(5 \times 10^5) =$ **2.0×10^{-6} m**

(b) $d \sin\theta = n\lambda$. $\sin\theta = n\lambda/d = 2 \times 590 \times 10^{-9} / 2.0 \times 10^{-6} = 0.590$

$\theta = \sin^{-1}(0.590) =$ **36.2 degrees**

(c) Max order: $n_{max} = d/\lambda = 2.0 \times 10^{-6} / 590 \times 10^{-9} = 3.39$, so $n_{max} =$ **3** (rounded down)

Exam Tip: The maximum order observable is found by setting sinθ = 1 (i.e. θ = 90 degrees): $n_{max} = d/\lambda$, rounded down to the nearest integer. For each order, there is a maximum on both sides of the central maximum, so the total number of maxima = $2n_{max} + 1$ (including the zero order).

5.6 Refraction and Total Internal Reflection

Refraction: The change in direction of a wave as it passes between media with different wave speeds, caused by a change in velocity. The frequency stays constant; the wavelength changes.

Snell's Law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n = c / v$$

where n = refractive index (dimensionless), c = speed of light in vacuum, v = speed of light in the medium. When light enters a denser medium (higher n), it slows down and bends towards the normal.

Total Internal Reflection (TIR)

Occurs when light travels from a denser to a less dense medium and the angle of incidence exceeds the critical angle:

$$\sin \theta_c = n_2 / n_1$$

Conditions: (1) light goes from higher n to lower n; (2) angle of incidence > critical angle.

Worked Example: Calculate the critical angle for glass (n = 1.50) to air (n = 1.00).

Solution:

$\sin \theta_c = n_2/n_1 = 1.00/1.50 = 0.667$

$\theta_c = \sin^{-1}(0.667) =$ **41.8 degrees**

5.7 Lenses

Converging (Convex) Lens

Converges parallel rays to the principal focus (F). Focal length f is positive. Can form real or virtual images.

Diverging (Concave) Lens

Diverges rays so they appear to come from a virtual focus. Focal length f is negative. Always produces virtual, upright, diminished images.

Lens Equation

$$1/f = 1/v + 1/u$$

Sign convention: real is positive, virtual is negative for image distance.

Magnification and Power

$$\text{Magnification} = v/u = \text{image height / object height}$$

$$\text{Power } P = 1/f \text{ (dioptres, D; f in metres)}$$

Converging lenses have positive power; diverging lenses have negative power.

5.8 Electromagnetic Spectrum

All EM waves travel at $c = 3.00 \times 10^8$ m s^{-1} in vacuum, are transverse, and need no medium.

Type	Wavelength	Uses
Radio	> 0.1 m	Broadcasting, communications
Microwave	0.1 m - 1 mm	Cooking, satellite comms, radar
Infrared	1 mm - 700 nm	Heating, thermal imaging, remote controls, optical fibres
Visible	700 - 400 nm	Vision, photography, optical fibres
Ultraviolet	400 - 10 nm	Sterilisation, fluorescence, tanning
X-rays	10 - 0.01 nm	Medical imaging, security scanning, crystallography
Gamma	< 0.01 nm	Cancer treatment, sterilisation, PET scans

5.9 Photoelectric Effect

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The **photoelectric effect** is the emission of electrons from a metal surface when EM radiation of sufficiently high frequency is incident. It provides key evidence for the particle (photon) nature of light.

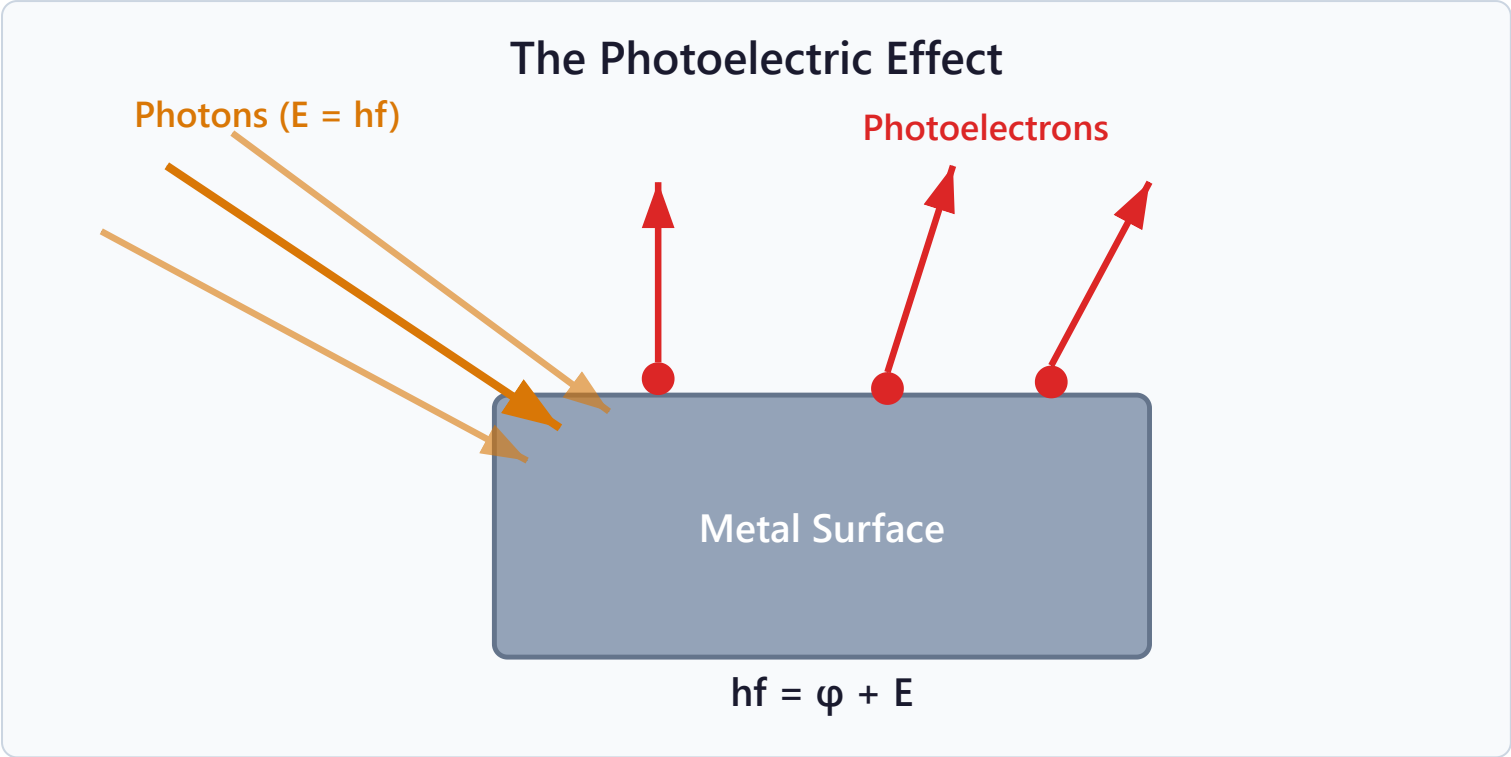
Key Observations (that wave theory cannot explain)

- Emission is **instantaneous** (no time delay, even at low intensity).
- Below a **threshold frequency f_0**, no electrons are emitted regardless of intensity.
- Increasing intensity increases the **number** of photoelectrons, not their maximum KE.
- Increasing frequency increases the **maximum KE** of the emitted electrons.

Einstein's Photoelectric Equation

$$hf = \varphi + E_{k(max)}$$

- h = Planck's constant = 6.63×10^{-34} J s
- f = frequency of incident photon
- φ = work function = minimum energy to release an electron = hf_0
- $E_{k(max)}$ = maximum kinetic energy of emitted photoelectron = $\frac{1}{2}mv^2_{max}$



k(max)

Worked Example: The work function of sodium is 2.28 eV. UV light of wavelength 250 nm is incident on a sodium surface. Calculate (a) the energy of each photon, (b) the maximum KE of the emitted electrons, (c) the maximum speed of the electrons.

Solution:

(a) $E = hc/\lambda = (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (250 \times 10^{-9}) = 7.96 \times 10^{-19}$ J

$= 7.96 \times 10^{-19} / 1.60 \times 10^{-19} =$ **4.97 eV**

(b) $E_{k(max)} = hf - \varphi = 4.97 - 2.28 =$ **2.69 eV** $= 4.30 \times 10^{-19}$ J

(c) $E_k = ½mv^2$. $v = \sqrt{(2E_k/m)} = \sqrt{(2 \times 4.30 \times 10^{-19} / 9.11 \times 10^{-31})} =$ **9.71×10^5 m s^{-1}**

Exam Tip: The photoelectric effect cannot be explained by the wave model of light. Waves would predict: (1) any frequency could cause emission given enough time/intensity, (2) there would be a time delay as energy accumulates, (3) higher intensity would give higher KE. All three predictions are wrong. The photon model explains everything: one photon transfers all its energy to one electron in a single interaction.

5.10 Wave-Particle Duality

Wave-particle duality: All matter and radiation exhibit both wave and particle properties. Light behaves as waves (diffraction, interference) and as particles (photoelectric effect). Electrons behave as particles (deflection by fields) and as waves (electron diffraction).

de Broglie Wavelength

$$\lambda = h / mv = h / p$$

Every moving particle has an associated wavelength. The faster or more massive the particle, the shorter its wavelength. Wave behaviour is observable when λ is comparable to the size of obstacles (e.g. electrons diffracted by atomic spacing ~10^{-10} m).

Worked Example: Calculate the de Broglie wavelength of an electron accelerated through a PD of 200 V.

Solution:

KE = eV = $1.60 \times 10^{-19} \times 200 = 3.20 \times 10^{-17}$ J

$\frac{1}{2}mv^2 = 3.20 \times 10^{-17}$, so $v = \sqrt{(2 \times 3.20 \times 10^{-17} / 9.11 \times 10^{-31})} = 8.38 \times 10^6$ m s^{-1}

$\lambda = h/mv = 6.63 \times 10^{-34} / (9.11 \times 10^{-31} \times 8.38 \times 10^6)$ = **8.67×10^{-11} m** (0.087 nm)

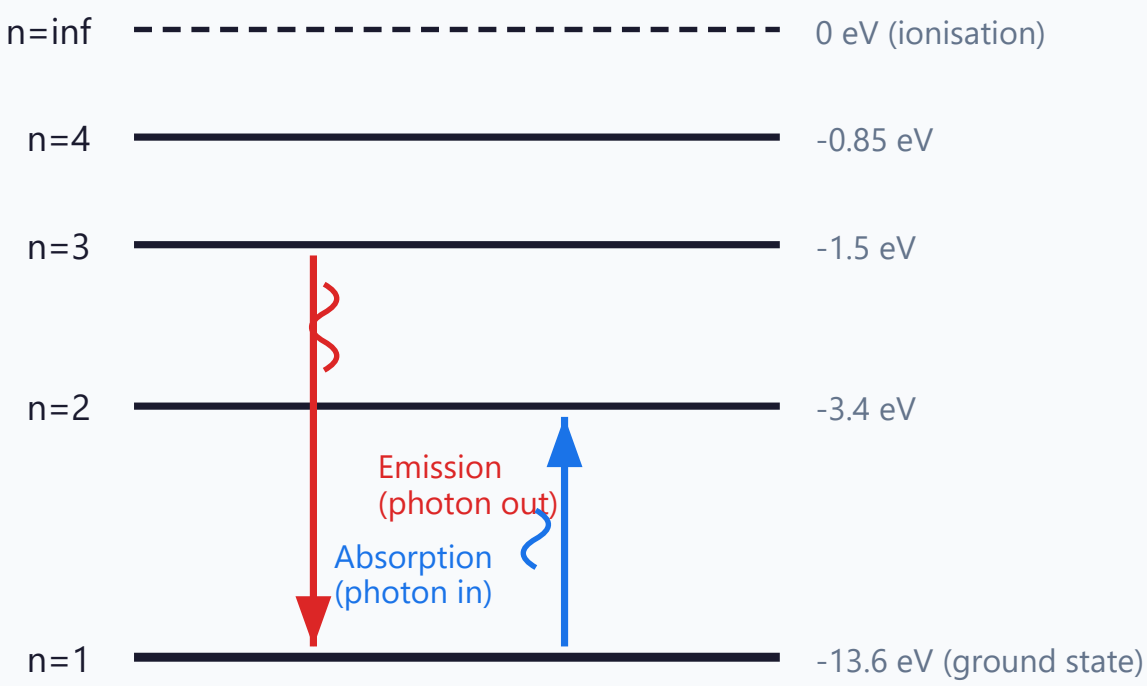
This is comparable to atomic spacing, so electron diffraction is observable.

5.11 Energy Levels and Photon Emission

Electrons in atoms exist in discrete **energy levels**. The ground state (n = 1) is the lowest. When an electron transitions between levels, it emits or absorbs a photon whose energy exactly equals the energy gap.

$$\Delta E = E_2 - E_1 = hf = hc/\lambda$$

Atomic Energy Levels and Transitions



$$\Delta E = hf = hc/\lambda$$

- **Emission:** Electron drops to a lower level, emitting a photon with E = ΔE between levels.
- **Absorption:** Electron absorbs a photon and jumps to a higher level. Photon energy must exactly match the gap.
- **Line spectra:** Discrete energy levels produce characteristic line spectra. Emission spectra show bright lines; absorption spectra show dark lines against a continuous background.

The Electronvolt

$$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$$

The energy gained by one electron accelerated through 1 V. Convenient for atomic-scale energies.

Exam Tip: Energy levels are negative (bound states). The ground state is the most negative. Convert eV to joules (multiply by 1.60×10^{-19}) before using E = hf to find frequency or wavelength. The ionisation energy from the ground state equals the magnitude of the ground state energy (13.6 eV for hydrogen).

Summary of Key Equations

Quantity	Equation
Wave equation	$v = f\lambda$
Double slit	$w = \lambda D/d$
Diffraction grating	$d \sin\theta = n\lambda$
Snell's law	$n_1 \sin\theta_1 = n_2 \sin\theta_2$
Critical angle	$\sin\theta_c = n_2/n_1$
Lens equation	$1/f = 1/v + 1/u$
Photon energy	$E = hf = hc/\lambda$
Photoelectric	$hf = \varphi + E_{k(max)}$
de Broglie	$\lambda = h/p = h/(mv)$
Energy levels	$\Delta E = hf$

⚗ Core Practical 2: Stationary waves on a string

Aim

To investigate how frequency, length, and tension affect stationary waves on a vibrating string. Verify $f = (1/2L)\sqrt{(T/\mu)}$.

Apparatus

Signal generator, vibration generator, string (known μ), pulley, slotted masses, metre ruler.

Method

1. Attach string to vibrator. Pass over pulley with hanging mass (tension T = mg).
2. Slowly increase frequency until the fundamental (1 antinode) forms. Record f and L.
3. Find higher harmonics. For each: $\lambda = 2L/n$.
4. Vary tension (change mass) and repeat. Plot f vs 1/L or f^2 vs T.

Safety

Safety screen in case string snaps. Cushion under masses. Clamp vibrator firmly.

Sources of Error

Error	Type	Improvement
Judging exact resonance	Random	Use paper rider (falls off at antinode at resonance)
String vibrates in wrong plane	Random	Ensure vibrator aligned; string horizontal
Frequency dial inaccurate	Systematic	Use digital frequency meter

Exam Tip: At the fundamental, $\lambda = 2L$. For the nth harmonic: $\lambda = 2L/n$, $f_n = nf_1$. The number of antinodes equals the harmonic number.

⚗ Core Practical 6: Sonometer investigation (length and tension effects on resonant frequency)

Aim

To investigate how the resonant frequency of a stretched wire depends on length and tension using a sonometer.

Apparatus

Sonometer (hollow wooden box with wire), two movable bridges, tuning fork(s) of known frequency (or microphone + CRO), mass hanger and masses, metre ruler, paper rider.

Method

1. Set tension T = mg using hanging mass. Place paper rider on wire at midpoint.
2. Strike tuning fork, press base on sonometer box.
3. Adjust bridge position until resonance (paper rider flies off). Measure vibrating length L.
4. **Effect of length:** Keep T constant, use different tuning forks, find resonant L each time. Plot f vs 1/L.
5. **Effect of tension:** Keep L constant, vary hanging mass, find resonance. Plot f^2 vs T.

Analysis

$f = (1/2L)\sqrt{(T/\mu)}$. Plot f vs 1/L: gradient = $(1/2)\sqrt{(T/\mu)}$. Plot f^2 vs T: gradient = $1/(4L^2\mu)$.

Safety

Goggles (wire under tension). Cushion under masses. Do not over-tighten wire.

Sources of Error

Error	Type	Improvement
Judging exact resonance point	Random	Paper rider method; try microphone + CRO for precise match
Measuring L to wrong bridge edge	Random	Measure from inner edge of each bridge; repeat
Friction at pulley reduces effective tension	Systematic	Use low-friction pulley

Exam Tip: The paper rider confirms resonance: maximum energy transfer from tuning fork to wire causes it to vibrate violently and throw off the rider. If the wire is not at resonance, the rider stays still.

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